Anshuman Pandey

Machine Learning Engineer

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Professional Summary

Final-year B.Tech CSE (AI/ML) student at VIT Bhopal focused on building production-grade machine learning systems, not just notebook prototypes. Experience spans native C/C++ ML engines (a deterministic resume-scoring engine, a from-scratch neural network trainer) and a neural chess engine combining deep reinforcement learning with tree search. Core stack is Python, PyTorch, and C/C++, with a growing focus on performance, reliability, and deployability of ML systems. Seeking a Machine Learning Engineer or AI Engineer internship, or an entry-level role.

Technical Skills

Languages	Python, C, C++, Java, SQL, HTML/CSS
ML / Deep Learning	PyTorch, TensorFlow, Scikit-learn, XGBoost, NumPy, Pandas, OpenCV, CNN, LSTM, ResNet, YOLOv8
LLMs & Retrieval	BM25, Sentence-Transformers (MiniLM), Hugging Face
Systems & Deployment	FastAPI, Flask, Docker, pybind11 / ctypes (Python↔C++ FFI), GitHub Actions (CI/CD), AWS
Concepts	Data Structures & Algorithms, MCTS, Reinforcement Learning

Projects

TalentMatch AI – Deterministic Resume Ranking Engine 2026
Python, C++, FastAPI, XGBoost *live demo*

- Owned the ML pipeline and C++ engine architecture (2-person team): led the v2 rewrite that replaced an LLM/Groq-based scorer with a fully deterministic native engine – Aho-Corasick trie skill matching, BM25 text retrieval, and ∼85 engineered features feeding an XGBoost ranker with a linear fallback.
- Developed the Python orchestration layer (PDF extraction, local MiniLM embeddings) and a ctypes FFI bridge connecting it to the C++ engine, exposed via FastAPI with a rule-based explanation module.

Nocturne – From-Scratch C/C++ Deep Learning Engine 2026
C, C++17, pybind11, Flask *nn-trainer.onrender.com*

- Engineered a neural network training system entirely in hand-written C/C++ – no PyTorch, TensorFlow, or scikit-learn – implementing forward/backward propagation, Xavier/He initialization, and SGD optimization at the primitive level.
- Bound the C/C++ core to Python with pybind11, releasing the GIL during training so a Flask server could stream live loss curves and weight matrices to the browser over Server-Sent Events, with an interactive weight editor for hand-editing values before training.

AlphaZ0 – Neural Chess Engine 2026
Python, PyTorch, C++17, MCTS, A3C *GitHub*

- Designed a ResNet dual-head network (10 residual blocks, 128 channels, policy and value heads) inspired by DeepMind's AlphaZero, layered on a custom rules engine and an 18-plane board encoder; search uses MCTS with PUCT selection and Dirichlet root noise, trained via a parallel A3C self-play loop with asynchronous GAE-advantage gradients.
- Rebuilt the performance-critical path (move generation, MCTS, self-play loop) natively in C++17 via pybind11 for v2, shipping a FastAPI backend that serves real-time bot moves to a browser UI.

Experience

Virtual Intern May 2026
ServiceNow *Remote*

- Completed a virtual internship in platform administration and automation, gaining hands-on experience across 4 core areas: Flow Designer, the Automated Test Framework (ATF), Reporting, and Agentic AI concepts.

Education

B.Tech in Computer Science & Engineering (AI/ML) 2023 – Present
Vellore Institute of Technology, Bhopal *CGPA: 8.54/10* Class XII: 89.5% | Class X: 89.4%

Certifications

Machine Learning (Coursera), AI & Agentic AI (TalentSprint), AWS Trainer & AWS Academy (AWS), C & C++ Programming (Udemy, HackerRank)

Achievements

2nd Runner-up, IEEE-NSUT Ideathon | Top 10, BITS Pilani Hackathon (national-level)